Seat No._____

GANPAT UNIVERSITY
B.Sc. IT (DS) SEM-I EXAMINATION

April - June 2023

U71A4IDS: INTRODUCTION TO DATA SCIENCE

Remedial

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. Discuss applications of data science.
2. What is regression? Why it is used? Also, discuss linear regression in detail.
3. What is data visualization? Explore different data visualization techniques.
4. What is data science? Explain the process of it.
5. Explain confusion matrix.
6. Which phenomena indicate that model is over fitting? Also, discuss the reasons behind it and enlist the solutions.
7. Define the following terms in context to the statement: "Data in the real world is dirty". Also, give an example to illustrate the meaning of the following terms.

   (1) Incomplete (2) Error (3) Outliers (4) Inconsistent

   (5) Duplicate (6) Noise

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Suppose you are given a survey data of employees and it has some missing values. So, how would you deal with the missing values from that survey?
2. What is data preprocessing? Discuss data cleaning in detail.
3. Distinguish between data science and business intelligence.

3. **Answer the following: (Any Two)** **(12)**

1. Explain stepwise regression.
2. Explain logistic regression.
3. Discuss the features and advantages of Python language in the field of data science.

4. **Answer the following: (Any One)** **(06)**

1. What is under fitting? Discuss the reasons and solutions to resolve problem of under fitting.
2. What is hyper parameter? Why it is important? Also, discuss any one technique to optimize the hyper parameter.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc.IT (DATA SCIENCE) SEM-I EXAMINATION

U71A5FM FUNDAMENTAL OF MATHEMATICS

Remedial April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer
4. Use of simple calculator is allowed

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. Let $U = \{1,2,3,\dots,15\}$,

   $A = \{x: x \text{ is a natural number and a factor of } 15\}$,

   $B = \{x: x \text{ is a natural number and less than } 6\}$.

   Find (i) $A \cup B$ (ii) $A \cap B$ (iii) A' and (iv) B'

2. There are 15 students in a group. Of these, 8 students are selected for only sports and 4 are selected for both sports and NCC. Assume that all the students selected in at least one of the activity.

   Find (i) The number of students selected for sports.

   (ii) The number of students selected for NCC.

   (iii) The number of students selected for NCC only

3. If $f(x) = \begin{cases} 2+5x\ , & x \geq 0 \\ 2-5x\ , & x < 0 \end{cases}$

   Then find $f(2) + f(0) + f(-2) + f(3) - f(5)$

4. Discuss the Homeorphic Graph in details.

5. Define the following terms in context of graph theory with suitable example:

   Graph, Simple graph, Pendent vertex, Closed Walk, Paths.

6. If $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & 0 \\ 0 & 2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & -2 \\ 8 & 4 & 4 \\ 4 & 2 & 1 \end{bmatrix}$ then find BA and AB.

7. How many different license plates can be made if each plat contains a sequence of two uppercase English letters followed by three digits?

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. If $A = \begin{bmatrix} 2 & 1 & 0 \\ 5 & 2 & -5 \\ 8 & -1 & 11 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & 3 \\ -7 & 5 & 45 \\ -5 & 2 & 10 \end{bmatrix}$ then find $A + B$ and $A - B$.

2. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 2 & 4 \\ -3 & 4 & 3 \end{bmatrix}$ then check whether the given matrix is symmetric or not? Justify your answer. Then evaluate AB if $B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

3. Define the following set by Roaster method

   (i)$A = \{x/x$ is a 2 digit natural such that the sum of its digits is $5\}$

   (ii)$B = \{x/x \in N, x^2 < 16\}$

   (iii)$C = \{x/\ x^2 - 6 = 0, x \epsilon R\}$

3. Answer the following: (Any Two) **(12)**

1. Determine whether the following functions are one-one or many one

   (i)$f: N \to N, f(x) = x + 7$ (ii) $f: R \to R, f(x) = |x| + 2$

2. Discuss application of graph theory in computer science.

3. State True or False

   (i)A function is onto when its range and codomain are equal

   (ii) Function cannot return more than one value at a time

   (iii) If $f: R \to R, f(x) = 1$ then given function is called constant function.

   (iv) If $f: N \to R, f(x) = x^2 - 5$ then domain of given function is R.

   (v) If $f: Z \to R, f(x) = x^2 - \frac{1}{3}x$ then codomain of given function is R.

   (vi) Range of function is subset of domain.

4. Answer the following: (Any One) **(06)**

1. Find the number of words, with or without meaning, that can be formed with the letters of the word 'CHAIR'.

2. How many different ways can gold, silver, and bronze medals be awarded in an Olympic event with 12 athletes competing?

END OF PAPER